

Formation Control of Heterogeneous Nonlinear Planar Mobile Robots

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Abstract—This paper addresses the problem of formation control for a team of three heterogeneous nonlinear planar mobile robots. Each agent is modeled by nonlinear affine dynamics, and the controlled output is the planar position. The proposed approach consists of two main stages: first, input-output feedback linearization is applied locally to each agent, reducing the heterogeneous nonlinear dynamics to a common double-integrator model; second, a distributed state-feedback formation controller is applied on the linearized system to drive the agents toward a desired equilateral triangular moving formation. The approach is validated through numerical simulations in MATLAB/Simulink across multiple scenarios, including different reference trajectories (constant-velocity, circular, and sinusoidal motion), two communication graph topologies, saturation of the physical inputs u_i , and nonlinear dynamics. Additionally, robustness with respect to imperfect model knowledge in the feedback-linearizing law and the effect of dynamic link failures on formation maintenance are investigated.

I. INTRODUCTION

In this paper, we consider a team of three heterogeneous planar mobile robots, each characterized by a distinct nonlinear model. Agent 1 features anisotropic nonlinear drag, Agent 2 exhibits a position-dependent apparent mass, and Agent 3 presents nonlinear coupling between the x and y dynamics. To address the heterogeneity of the agents, a well-established two-stage approach is adopted. First, input-output feedback linearization is applied locally to each agent, exploiting the structure of the nonlinear affine dynamics to cancel the nonlinearities and reduce each robot to a common double-integrator model. This decoupling step is exact when the system model is perfectly known, and allows the subsequent formation controller to be designed entirely in the linear domain. Despite the differences in the original dynamics, after local feedback linearization all agents are governed by the same second-order dynamics, enabling the design of a unified distributed state-feedback controller. The formation protocol drives the three agents to form and maintain an equilateral triangular configuration while tracking a common reference trajectory. The communication between agents is modeled as a directed graph, and two distinct topologies are considered and compared. The performance of the closed-loop system is evaluated through extensive MATLAB/Simulink simulations across multiple scenarios, including constant-velocity, circular, and sinusoidal reference trajectories. Additional analyses investigate the effect of actuator saturation on the physical inputs, the robustness of the feedback-linearizing law under imperfect model knowledge, and the impact of dynamic link failures on formation maintenance.

II. AGENT MODELS AND INPUT-OUTPUT FEEDBACK LINEARIZATION

A. General Framework

Each agent $i = 1, 2, 3$ moves on the plane and is described by nonlinear affine dynamics of the form:

$$\dot{p}_i = v_i, \quad \dot{v}_i = \phi_i(p_i, v_i) + B_i(p_i, v_i)u_i \quad (1)$$

where $p_i = [p_{x,i}, p_{y,i}]^\top \in \mathbb{R}^2$ is the planar position, $v_i = [v_{x,i}, v_{y,i}]^\top \in \mathbb{R}^2$ is the velocity, $u_i = [u_{x,i}, u_{y,i}]^\top \in \mathbb{R}^2$ is the physical input, ϕ_i is the drift term, and B_i is the input matrix. The measured and controlled output for each agent is the planar position:

$$y_i = \begin{bmatrix} p_{x,i} \\ p_{y,i} \end{bmatrix} \quad (2)$$

The goal of input-output feedback linearization is to find, for each agent, a state-dependent input transformation $u_i = B_i(p_i, v_i)^{-1}[\nu_i - \phi_i(p_i, v_i)]$ such that the closed-loop input-output behavior becomes that of a double integrator:

$$\ddot{y}_i = \nu_i \quad (3)$$

where $\nu_i \in \mathbb{R}^2$ is a new virtual input generated by the formation controller.

B. Relative Degree and Absence of Zero Dynamics

For all three agents, the vector relative degree is verified as follows. The first time derivative of the output is:

$$\dot{y}_i = \begin{bmatrix} v_{x,i} \\ v_{y,i} \end{bmatrix} \quad (4)$$

which does not depend on the input u_i . The second derivative is:

$$\ddot{y}_i = \phi_i(p_i, v_i) + B_i(p_i, v_i)u_i \quad (5)$$

where the input appears explicitly through the decoupling matrix B_i . Therefore, the vector relative degree is $(r_x, r_y) = (2, 2)$, with $r_x + r_y = 4$. Since the state dimension of each agent is $\dim(x_i) = 4$, the feedback linearization is exact and complete: there are no zero dynamics.

The feedback-linearizing law is applicable provided the decoupling matrix $B_i(p_i, v_i)$ is nonsingular, i.e., $\det B_i(p_i, v_i) \neq 0$. This condition is verified individually for each agent below.

C. Agent 1: Planar Robot with Anisotropic Nonlinear Drag

The dynamics of Agent 1 are:

$$\dot{p}_x = v_x, \quad \dot{v}_x = -0.35v_x - 0.08v_x|v_x| + \frac{1}{1.2}u_x \quad (6)$$

$$\dot{p}_y = v_y, \quad \dot{v}_y = -0.45v_y - 0.10v_y|v_y| + \frac{1}{1.2}u_y \quad (7)$$

The drift term and decoupling matrix are identified as:

$$\phi_1 = \begin{bmatrix} -0.35v_x - 0.08v_x|v_x| \\ -0.45v_y - 0.10v_y|v_y| \end{bmatrix}, \quad B_1 = \begin{bmatrix} \frac{1}{1.2} & 0 \\ 0 & \frac{1}{1.2} \end{bmatrix} \quad (8)$$

Since B_1 is diagonal with strictly positive entries, it is nonsingular everywhere: $\det B_1 = \frac{1}{1.44} \neq 0$. The feedback-linearizing law is:

$$u_1 = B_1^{-1}[\nu_1 - \phi_1] = \begin{bmatrix} 1.2(\nu_{x,1} + 0.35v_x + 0.08v_x|v_x|) \\ 1.2(\nu_{y,1} + 0.45v_y + 0.10v_y|v_y|) \end{bmatrix} \quad (9)$$

D. Agent 2: Planar Robot with Position-Dependent Apparent Mass

The dynamics of Agent 2 are:

$$\dot{p}_x = v_x, \quad \dot{v}_x = -0.40v_x + \frac{1}{1 + 0.25\sin^2(p_x)}u_x \quad (10)$$

$$\dot{p}_y = v_y, \quad \dot{v}_y = -0.55v_y + \frac{1}{1 + 0.30\cos^2(p_y)}u_y \quad (11)$$

The drift term and decoupling matrix are:

$$\phi_2 = \begin{bmatrix} -0.40v_x \\ -0.55v_y \end{bmatrix}, \quad B_2 = \begin{bmatrix} \frac{1}{1+0.25\sin^2(p_x)} & 0 \\ 0 & \frac{1}{1+0.30\cos^2(p_y)} \end{bmatrix} \quad (12)$$

Since $\sin^2(\cdot) \geq 0$ and $\cos^2(\cdot) \geq 0$, both diagonal entries of B_2 are strictly positive for all (p_i, v_i) , hence $\det B_2 > 0$ everywhere. The feedback-linearizing law is:

$$u_2 = B_2^{-1}[\nu_2 - \phi_2] = \begin{bmatrix} (1 + 0.25\sin^2(p_x))(\nu_{x,2} + 0.40v_x) \\ (1 + 0.30\cos^2(p_y))(\nu_{y,2} + 0.55v_y) \end{bmatrix} \quad (13)$$

E. Agent 3: Planar Robot with Nonlinear Coupling

The dynamics of Agent 3 are:

$$\dot{p}_x = v_x, \quad \dot{v}_x = -0.30v_x + 0.15\sin(p_y)v_y + (1 + 0.20\cos^2(p_y))u_x + 0.08\sin(p_x)u_y \quad (14)$$

$$\dot{p}_y = v_y, \quad \dot{v}_y = -0.35v_y + 0.15\cos(p_x)v_x + 0.08\sin(p_y)u_x + (1 + 0.20\sin^2(p_x))u_y \quad (15)$$

This agent presents nonlinear coupling between the x and y channels both in the drift and in the input matrix. The drift term and decoupling matrix are:

$$\phi_3 = \begin{bmatrix} -0.30v_x + 0.15\sin(p_y)v_y \\ -0.35v_y + 0.15\cos(p_x)v_x \end{bmatrix} \quad (16)$$

$$B_3 = \begin{bmatrix} 1 + 0.20\cos^2(p_y) & 0.08\sin(p_x) \\ 0.08\sin(p_y) & 1 + 0.20\sin^2(p_x) \end{bmatrix} \quad (17)$$

The determinant of B_3 is:

$$\det B_3 = (1 + 0.20\cos^2(p_y))(1 + 0.20\sin^2(p_x)) - 0.0064\sin(p_x)\sin(p_y) \quad (18)$$

Since the diagonal terms satisfy $1 \leq 1 + 0.20(\cdot) \leq 1.20$, their product is at least 1. The off-diagonal product $|0.0064\sin(p_x)\sin(p_y)| \leq 0.0064 \ll 1$, therefore $\det B_3 > 0$ for all states, confirming nonsingularity. The feedback-linearizing law is:

$$u_3 = B_3(p_3, v_3)^{-1}[\nu_3 - \phi_3] \quad (19)$$

F. Linearized Model and Virtual Input

After local feedback linearization, all three agents satisfy the common double-integrator dynamics:

$$\ddot{y}_i = \nu_i, \quad i = 1, 2, 3 \quad (20)$$

Equivalently, defining $w_i = \dot{y}_i$:

$$\dot{y}_i = w_i, \quad \dot{w}_i = \nu_i \quad (21)$$

The formation controller is designed entirely at the level of the virtual input ν_i . The physical input u_i applied to each nonlinear robot is then recovered locally via the feedback-linearizing law derived above. The state of the linearized system for each agent is $x_i = [p_{x,i}, v_{x,i}, p_{y,i}, v_{y,i}]^\top \in \mathbb{R}^4$, governed by:

$$\dot{x}_i = Ax_i + B\nu_i \quad (22)$$

where:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \quad (23)$$

which correspond exactly to the matrices used in the MATLAB implementation.

III. DISTRIBUTED FORMATION CONTROLLER DESIGN

A. Formation Task and Feasibility

The three agents are required to form and maintain an equilateral triangular configuration while tracking a common reference trajectory $r(t) \in \mathbb{R}^2$. The desired formation offsets are defined as:

$$h_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad h_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad h_3 = \begin{bmatrix} 0.5 \\ \sqrt{3}/2 \end{bmatrix} \quad (24)$$

so that the desired trajectory for each agent is $y_i^*(t) = r(t) + h_i - h_1$. In the state-space of the linearized double-integrator model, the formation offset for agent i is extended to:

$$\tilde{h}_i = \begin{bmatrix} h_{x,i} \\ 0 \\ h_{y,i} \\ 0 \end{bmatrix} \in \mathbb{R}^4 \quad (25)$$

For the formation to be feasible, the desired offset must be compatible with the system dynamics, i.e., it must satisfy:

$$A(\tilde{h}_i - \tilde{h}_1) = 0, \quad i = 2, 3 \quad (26)$$

Since $\tilde{h}_i - \tilde{h}_1$ has zero velocity components, and A selects exactly those components, the product $A(\tilde{h}_i - \tilde{h}_1) = 0$ trivially, confirming that the triangular formation is feasible.

B. Communication Graph

The interaction among agents is modeled as a directed weighted graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{1, 2, 3\}$ is the set of nodes, $\mathcal{E}$ is the set of directed edges. Two distinct topologies are considered and compared:

- **Topology 1 (chain):** $1 \rightarrow 2 \rightarrow 3$, where agent 2 receives information from agent 1 with weight $a_{21} = 3$, and agent 3 receives from agent 2 with weight $a_{32} = 2$.
- **Topology 2 (star):** $3 \leftarrow 1 \rightarrow 2$, where both agents 2 and 3 receive directly from agent 1, with weights $a_{21} = 3$ and $a_{31} = 2$.

Both topologies contain a directed spanning tree rooted at agent 1, which acts as the leader and is the only node with a local tracking loop on the reference trajectory $r(t)$. The feedforward term $u_{ff}(t)$, however, is broadcast to all agents to ensure that the entire formation translates rigidly. The Laplacian matrix $L = D - A_{adj}$ and the leader pinning matrix $\Pi_1 = \text{diag}(1, 0, 0)$ are computed accordingly for each topology.

C. State-Feedback Formation Protocol

After feedback linearization, each agent is governed by the double-integrator dynamics $\dot{x}_i = Ax_i + Bv_i$. The distributed state-feedback formation controller for agent i is:

$$v_i = cK\varepsilon_i + u_{ff}(t) + \pi_i u_{fb}(t) \quad (27)$$

where:

- $\varepsilon_i = \sum_{j \in \mathcal{N}_i} a_{ij} [(x_j - \tilde{h}_j) - (x_i - \tilde{h}_i)]$ is the neighborhood formation error, measuring how far agent i deviates from its desired relative configuration with respect to its neighbors;
- $u_{ff}(t) = \ddot{r}(t)$ is the feedforward term, broadcast to all agents, ensuring that the entire formation translates rigidly along the reference trajectory;
- $u_{fb}(t) = K_\ell(x_r(t) - x_1(t))$ is the local tracking feedback term, weighted by π_i , the i -th diagonal entry of $\Pi_1 = \text{diag}(1, 0, 0)$, which activates this term exclusively for the leader node (agent 1), enforcing tracking of the reference trajectory $r(t)$. The matrix $K_\ell = c_\ell K$ shares the same structure as the formation gain, where $c_\ell > 0$ is a scalar that tunes the tracking speed of the leader independently from the coupling gain c . In the nominal configuration, $c_\ell = c$ is adopted, so that a uniform weighting is applied across formation and tracking; the effect of increasing c_ℓ is investigated in Section V.
- $c > 0$ is the coupling gain, selected to ensure closed-loop stability as detailed in Section III-E;
- $K \in \mathbb{R}^{2 \times 4}$ is the feedback gain matrix designed via LQR, as described in Section III-D.

The resulting control laws for the leader and the followers are therefore:

$$v_1 = cK\varepsilon_1 + u_{ff}(t) + u_{fb}(t), \quad v_i = cK\varepsilon_i + u_{ff}(t), \quad i = 2, 3 \quad (28)$$

The physical input u_i applied to each nonlinear robot is then recovered locally via the feedback-linearizing law:

$$u_i = B_i(p_i, v_i)^{-1} [\nu_i - \phi_i(p_i, v_i)] \quad (29)$$

D. FeedBack Gain Matrix Selection

The feedback gain matrix K is computed via Linear Quadratic Regulator (LQR) theory by solving the Algebraic Riccati Equation (ARE):

$$A^\top P + PA - PBR^{-1}B^\top P + Q = 0 \quad (30)$$

and setting:

$$K = R^{-1}B^\top P \quad (31)$$

The weighting matrices are chosen as $Q = 10 \cdot I_4$ and $R = I_2$, reflecting a design choice that penalizes state deviations more heavily than control effort.

E. Coupling Gain Selection

The stability of the closed-loop multi-agent system requires the coupling gain c to satisfy:

$$c > c_{\min} = \frac{1}{2 \min_i \text{Re}(\lambda_i(L + \Pi_1))} \quad (32)$$

where $\lambda_i(L + \Pi_1)$ denotes the eigenvalues of the matrix $L + \Pi_1$. To ensure a sufficient stability margin, the coupling gain is set to $c = 3 c_{\min}$. The leader gain is set as $K_\ell = c_\ell K$, where c_ℓ is chosen independently from c , allowing to tune separately the rigidity of the formation and the tracking speed of the leader node. In the nominal configuration, $c_\ell = c$ is used.

F. Reference Trajectories

Three reference trajectories $r(t)$ are considered and implemented in the MATLAB/Simulink simulation environment:

- **Constant velocity:** $r(t) = v_0 t$, producing straight-line motion.
- **Circular:** $r(t) = [\rho \cos(\omega t), \rho \sin(\omega t)]^\top$, producing periodic curved motion.
- **Sinusoidal:** $r(t) = [v_0 t, A \sin(\omega t)]^\top$, combining forward motion with lateral oscillations.

Although all three trajectories are available in the simulation environment, the quantitative analysis presented in this report focuses on the circular trajectory. This choice is motivated by the fact that circular motion is the most representative scenario for assessing the full capabilities of the controller: unlike constant velocity motion, where $\ddot{r}(t) = 0$ trivially cancels the feedforward contribution, the circular trajectory produces a non-zero and continuously varying reference acceleration, which simultaneously exercises both the feedforward term $u_{ff}(t)$ and the feedback correction.

IV. SIMULATION RESULTS

A. Simulation Setup

All simulations are carried out in MATLAB/Simulink over a time horizon of $T = 100$ s, using a circular reference trajectory $r(t) = [\rho \cos(\omega t), \rho \sin(\omega t)]^\top$ with radius $\rho = 5$ m and angular frequency $\omega = 0.5$ rad/s. The controller parameters are set as described in Sections III-D and III-E. The initial conditions of the three agents are set as:

$$x_1(0) = \begin{bmatrix} 3 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad x_2(0) = \begin{bmatrix} -4 \\ 0 \\ 7 \\ 0 \end{bmatrix}, \quad x_3(0) = \begin{bmatrix} -2 \\ 0 \\ -4 \\ 0 \end{bmatrix} \quad (33)$$

where each state vector is organized as $x_i = [p_{x,i}, v_{x,i}, p_{y,i}, v_{y,i}]^\top$. These values are fixed across all scenarios to ensure a consistent and reproducible baseline for comparison; the simulation script also supports randomly generated initial conditions. Nine simulation scenarios are analyzed, covering two communication topologies (chain and star), two agent models (linearized and nonlinear), actuator saturation, dynamic link failures, and imperfect model knowledge in the feedback-linearizing law.

B. Scenario 1: Linearized Model, Chain Topology

Figures 1–4 show the results for the nominal case: linearized agent models with chain topology $1 \rightarrow 2 \rightarrow 3$. The XY trajectory plot confirms that all three robots converge to the desired circular formation: starting from separated initial positions, they progressively align into the equilateral triangular configuration, which is maintained throughout the simulation. The formation errors along both axes decay smoothly to zero within approximately 10 s, with no oscillations, confirming the effectiveness of the controller. The physical inputs exhibit large transient peaks at $t = 0$ due to the initial position mismatch, and then settle to small periodic values dictated by the circular acceleration of the reference trajectory.

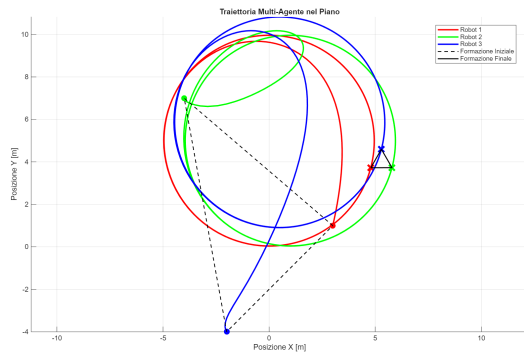


Fig. 1. Scenario1-XY trajectory:linearized model, chain topology.

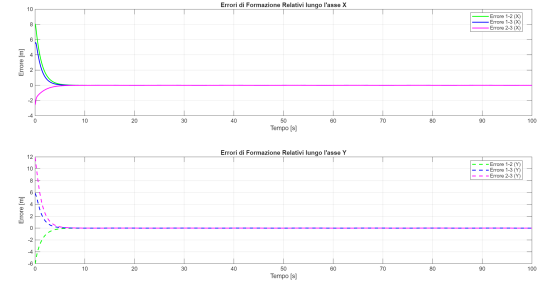


Fig. 2. Scenario1-Formation errors:linearized model, chain topology.

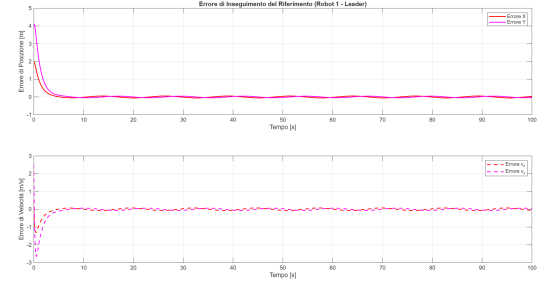


Fig. 3. Scenario1-Leader tracking error:linearized model, chain topology

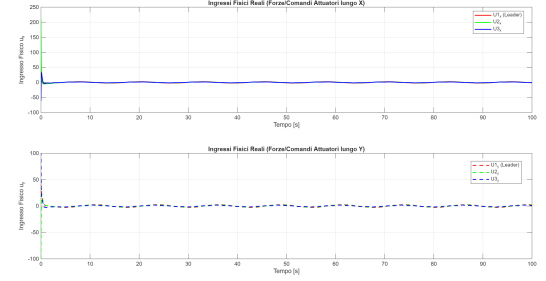


Fig. 4. Scenario1-Physical inputs:linearized model, chain topology.

C. Scenario 2: Linearized Model, Star Topology

The star topology $3 \leftarrow 1 \rightarrow 2$ produces results that are numerically identical to those of Scenario 1. This is not a coincidence: despite the structural difference between the two topologies, the matrices $L + \Pi_1$ share the same diagonal entries, yielding the same eigenvalues $\lambda = \{1, 2, 3\}$. As a consequence, the minimum coupling gain $c_{\min} = 0.5$ are identical for both topologies. Since the gain matrix K , the coupling gain c , and the initial conditions are all shared, the closed-loop dynamics are governed by exactly the same matrices, and the XY trajectories, formation errors, and physical inputs are indistinguishable from those of Scenario 1. For this reason, no additional figures are reported here. A broader discussion of the structural differences between the two topologies and their implications is provided in Section V.

D. Scenario 3: Linearized Model, Chain Topology, Actuator Saturation

Figures 5–7 show the results when actuator saturation is imposed on the physical inputs u_i , with bounds $u_{\max} = 2$ and $u_{\min} = -2$. The effect of saturation is clearly visible:

the XY trajectories show more irregular transient paths in the initial phase, as the robots are unable to generate the large correction forces required. The formation errors exhibit a more pronounced overshoot and a slower convergence compared to the unsaturated case, reaching zero only after approximately 15 s. In steady state, the physical inputs oscillate periodically between the saturation bounds, driven by the circular acceleration of the reference, revealing that the required steady-state effort exceeds the actuator limits.

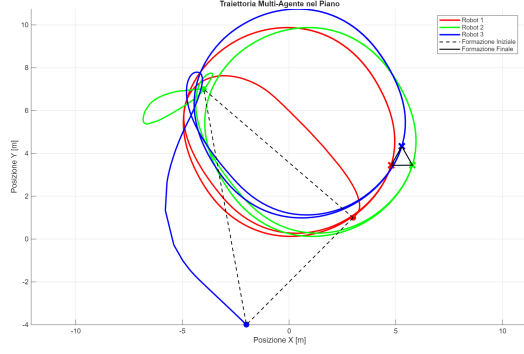


Fig. 5. Scenario3-XY trajectory:linearized model, chain topology, saturation.

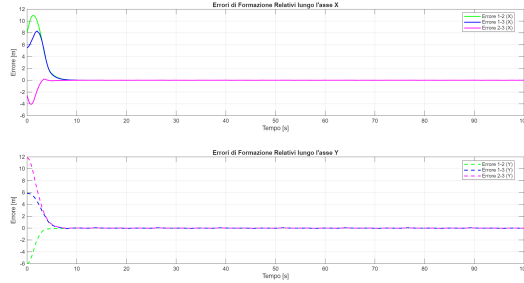


Fig. 6. Scenario3-Formation errors:linearized model, chain topology, saturation.

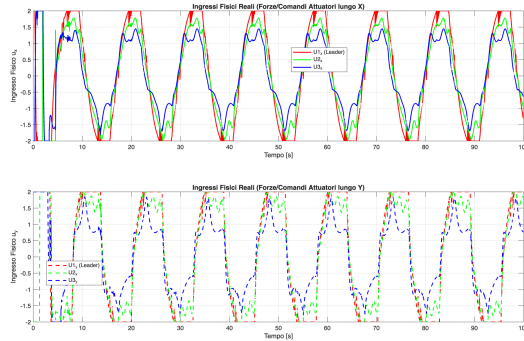


Fig. 7. Scenario3-Physical inputs:linearized model, chain topology, saturation.

E. Scenario 4: Nonlinear Model, Chain Topology

Figures 8–10 show the results when the full nonlinear dynamics are simulated, applying the virtual input ν_i directly to each nonlinear robot without invoking the feedback-linearizing law. Despite operating on the nonlinear model, the formation

is successfully achieved and maintained. The XY trajectories and formation errors are visually very similar to the linearized case, with convergence occurring within approximately 10 s. A slight residual oscillation is visible in the steady-state formation errors, attributable to the nonlinear perturbation terms ϕ_i that are not cancelled. The physical inputs are also comparable in magnitude, confirming the robustness argument: since $B_i \approx I$ and ϕ_i is dissipative, the controller designed on the linearized model retains its effectiveness on the nonlinear plant.

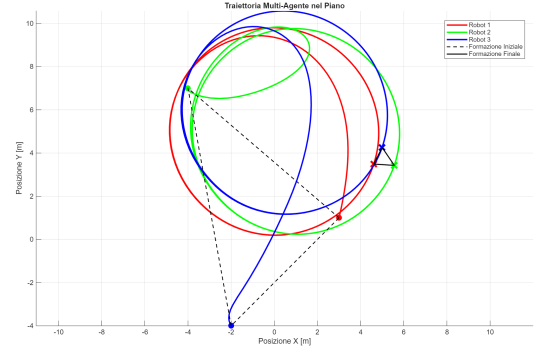


Fig. 8. Scenario4-XY trajectory:nonlinear model, chain topology.

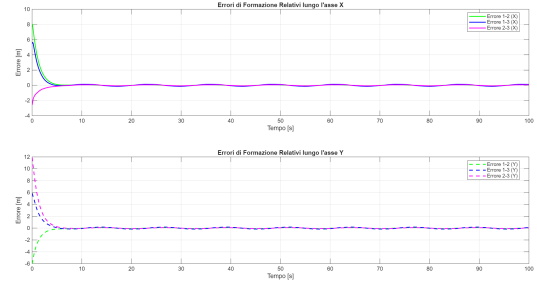


Fig. 9. Scenario4-Formation errors:nonlinear model, chain topology.

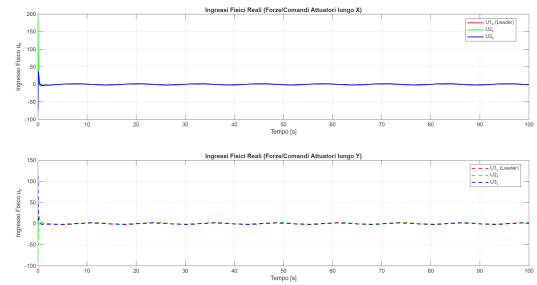


Fig. 10. Scenario4-Physical inputs:nonlinear model, chain topology.

F. Scenario 5: Nonlinear Model, Star Topology

As observed for the linearized case (Scenarios 1 and 2), the nonlinear model under the star topology produces results that are numerically identical to those of Scenario 4. The same argument applies: the matrices $L + \Pi_1$ for the two topologies share the same eigenvalues $\lambda = \{1, 2, 3\}$, yielding the same coupling gain $c = 1.5$. Since all other parameters

and initial conditions are unchanged, the XY trajectories, formation errors, and physical inputs are indistinguishable from those of Scenario 4 (Figures 8–10). For this reason, no additional figures are reported here.

G. Scenario 6: Nonlinear Model, Chain Topology, Actuator Saturation

Figures 11–13 show the combined effect of nonlinear dynamics and actuator saturation. As expected, the transient is more irregular than in all previous scenarios: the XY trajectories exhibit wider deviations from the desired circular path in the initial phase. The formation errors show a larger overshoot and a longer convergence time. In steady state, as in Scenario 3, the physical inputs saturate periodically in response to the circular acceleration demand. Despite this degradation, the formation is nonetheless achieved, confirming the resilience of the controller.

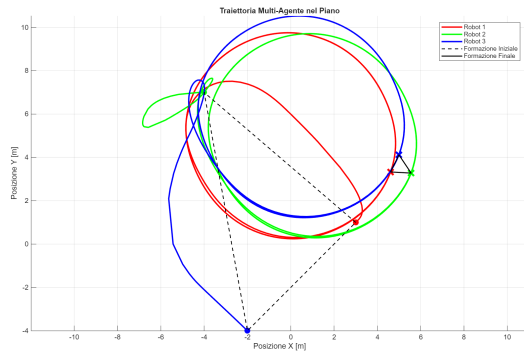


Fig. 11. Scenario6-XY trajectory:nonlinear model, chain topology, saturation.

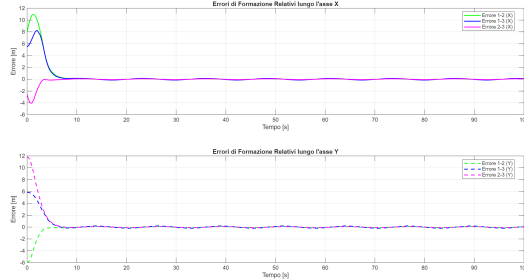


Fig. 12. Scenario6-Formation errors:nonlinear model, chain topology, saturation.

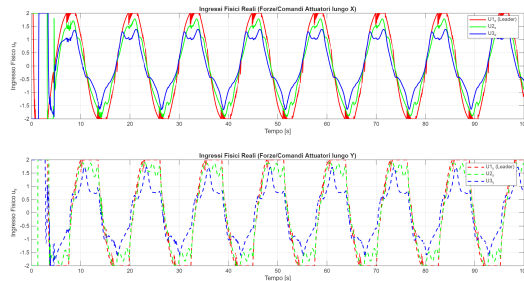


Fig. 13. Scenario6-Physical inputs:nonlinear model, chain topology, saturation.

H. Scenario 7: Effect of Asymmetric Leader Gain ($c_\ell = 1.5c$)

Figure 14 shows the results of the linearized model with chain topology when the leader gain is set to $c_\ell = 1.5c$, decoupling the tracking aggressiveness of agent 1 from the formation coupling gain. The XY trajectory, formation errors, and physical inputs are visually indistinguishable from those of Scenario 1, where $c_\ell = c$ was adopted. The leader tracking error, shown in Figure 14, confirms that agent 1 converges to the reference trajectory within approximately 5 s, with position errors below 0.1 m in steady state and velocity errors negligible after the initial transient. These results suggest that, for the chosen system parameters, the closed-loop performance is largely insensitive to moderate variations of c_ℓ around c , provided that c_ℓ remains within a safe range. A more detailed discussion is provided in Section V.

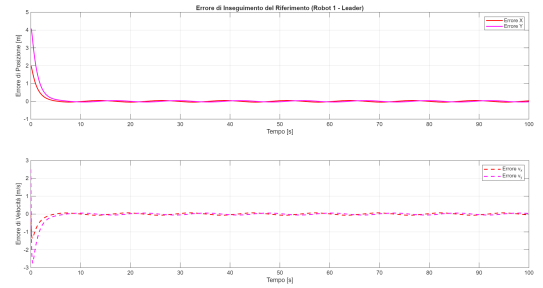


Fig. 14. Scenario7-Leader tracking error:linearized model, chain topology, $c_\ell = 1.5c$.

I. Scenario 8: Robustness to Imperfect Model Knowledge

Figures 15–17 show the behavior when the feedback-linearizing law is computed using a perturbed version of the agent models, simulating the realistic case where the controller has imperfect knowledge of the robot dynamics. Despite the model mismatch, the formation is achieved and maintained with performance visually indistinguishable from the nominal case (Scenario 1). The formation errors converge to zero within the same time window, and the physical inputs exhibit the same qualitative profile. This result confirms the inherent robustness of the controller to moderate model uncertainties in the feedback-linearizing law.

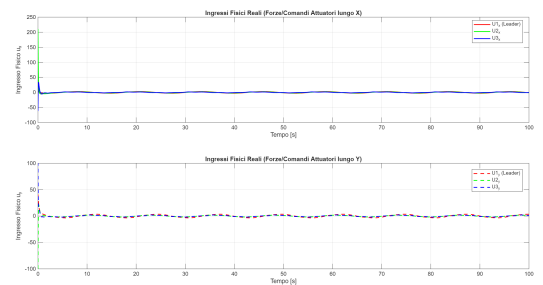


Fig. 15. Scenario8-XY trajectory:imperfect model knowledge.

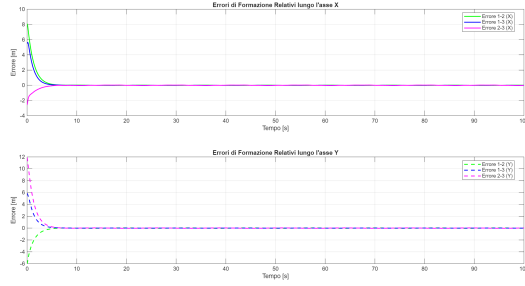


Fig. 16. Scenario 8-Formation errors:imperfect model knowledge.

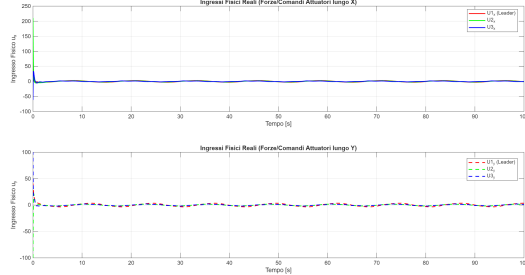


Fig. 17. Scenario 8-Physical inputs:imperfect model knowledge.

J. Scenario 9: Dynamic Link Failures

Figures 18–20 show the behavior of the linearized system under dynamic changes in the communication graph. Starting from the chain topology, the link $3 \rightarrow 2$ is removed at $t = 20$ s and the link $2 \rightarrow 3$ is removed at $t = 35$ s, leaving agent 3 completely isolated from the network. As a consequence, agent 3 loses all information about the reference and its neighbors: the formation error between agents 2 and 3 starts growing monotonically after $t = 35$ s, as visible in the XY trajectory where robot 3 drifts away from the desired orbit. Agents 1 and 2, which retain their communication link through the reference, maintain correct formation. This result highlights a fundamental limitation of distributed controllers: the loss of a spanning tree in the communication graph inevitably leads to loss of formation for the disconnected agents.

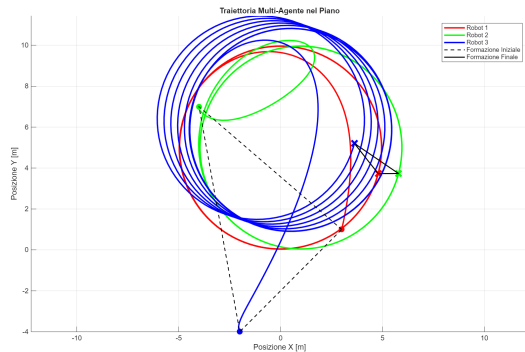


Fig. 18. Scenario 9-XY trajectory:dynamic link failures.

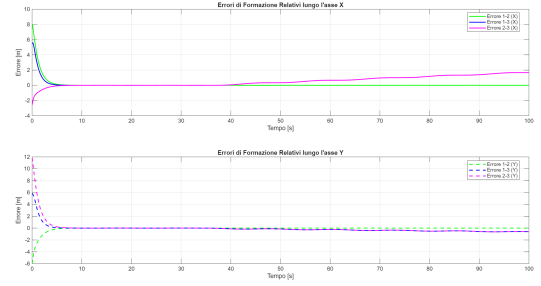


Fig. 19. Scenario 9-Formation errors:dynamic link failures.

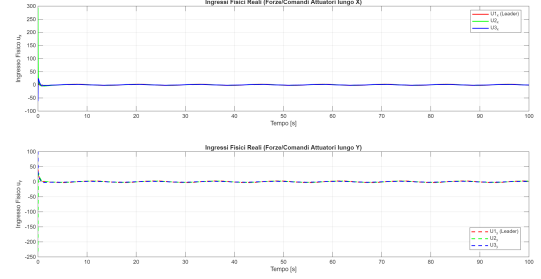


Fig. 20. Scenario 9-Physical inputs:dynamic link failures.

V. COMPARATIVE ANALYSIS

A. Chain Topology vs. Star Topology

Comparing Scenarios 1 and 2 (linearized model) and Scenarios 4 and 5 (nonlinear model), the two communication topologies yield virtually identical formation performance in terms of convergence time and steady-state error. This result is consistent with the theoretical condition: both topologies contain a directed spanning tree rooted at agent 1, which is the necessary and sufficient condition for formation convergence under the Distributed State Feedback protocol. The key structural difference is that in the chain topology, agent 3 receives information indirectly through agent 2, introducing a cascade dependency, whereas in the star topology, agents 2 and 3 both receive information directly from agent 1, resulting in a more symmetric information flow. Despite this structural difference, the eigenvalues of $L + \Pi_1$ are the same in both cases, yielding identical coupling gains c_{min} and therefore comparable closed-loop dynamics.

B. Linearized Model vs. Nonlinear Model

Comparing Scenarios 1 and 4 (chain topology, no saturation), the linearized and nonlinear models produce nearly identical results. The XY trajectories are visually indistinguishable, the formation errors converge within the same time window of approximately 10 s, and the physical inputs have the same order of magnitude. The only perceptible difference is a slight residual oscillation in the steady-state formation errors of the nonlinear case, caused by the uncompensated drift terms ϕ_i . This confirms the theoretical argument: since the decoupling matrices B_i are close to the identity and the drift terms ϕ_i are dissipative, the controller designed on the linearized model is inherently robust to the nonlinear perturbations introduced

by the real dynamics. The same conclusion holds for the star topology when comparing Scenarios 2 and 5.

C. Effect of Actuator Saturation

Comparing Scenarios 1 and 3 (linearized model, chain topology), the introduction of actuator saturation at $u_{max} = \pm 2$ has two distinct effects. During the transient phase, the saturation limits the corrective forces available to the robots, causing larger deviations in the XY trajectories and a delayed convergence of the formation errors, with overshoot reaching up to 11 m in the X direction compared to 8 m in the unsaturated case. More significantly, in steady state the saturation is persistently active: the circular reference trajectory requires a centripetal acceleration that exceeds the actuator bounds, causing the physical inputs to clip at ± 2 throughout the simulation. As a consequence, a small but non-zero residual formation error persists in steady state. This behavior is further amplified in Scenario 6 (nonlinear model with saturation), where the combination of nonlinear model and saturated inputs produces the most degraded transient among all scenarios, though formation convergence is still achieved.

D. Effect of Asymmetric Leader Gain

Comparing Scenario 1 ($c_\ell = c$) and Scenario 7 ($c_\ell = 1.5c$), the increase of the leader gain by a factor of 1.5 produces no appreciable difference in the closed-loop behavior. The formation errors, XY trajectories, and physical inputs are virtually identical in both scenarios. The leader tracking error in Scenario 7 confirms that agent 1 successfully follows the circular reference, with residual position oscillations below 0.1 m in steady state, consistent with the periodic acceleration demand of the circular trajectory.

E. Robustness to Model Uncertainty

Scenario 8 demonstrates that the feedback-linearizing law is robust to imperfect model knowledge. Despite the parameter mismatch in the feedback-linearizing law, the overall closed-loop performance is essentially identical to Scenario 1. This robustness comes from the fact that the nonlinear terms incompletely cancelled by the imperfect feedback-linearizing law behave similarly to the uncompensated ϕ_i in the nonlinear model scenario (Scenario 4), which was already shown to have a negligible impact. This result has practical relevance: it suggests that in real deployments, where exact model knowledge is rarely available, the proposed two-stage architecture remains effective under moderate parametric uncertainty.

F. Effect of Dynamic Link Failures

Scenario 9 reveals the critical importance of maintaining a connected spanning tree in the communication graph. The sequential removal of links $3 \rightarrow 2$ at $t = 20$ s and $2 \rightarrow 3$ at $t = 35$ s progressively degrades the information available to agent 3. After $t = 35$ s, when agent 3 is fully disconnected, it loses all reference information and its formation error grows unboundedly, as clearly visible in the XY trajectory where

robot 3 drifts onto a different trajectory. Agents 1 and 2, which retain their mutual link and access to the reference, maintain correct formation. This scenario underlines that the robustness of the distributed controller is entirely conditioned on the connectivity of the communication graph: the controller is robust to model uncertainties and nonlinearities, but has no mechanism to compensate for the loss of topological connectivity.

VI. CONCLUSION

This paper presented the design and simulation of a distributed formation control system for a team of three heterogeneous nonlinear planar mobile robots. The proposed approach combined input-output feedback linearization, applied locally to each agent, with a distributed state-feedback controller designed on the resulting common double-integrator model. The feedback linearization step exactly cancelled the nonlinear dynamics of each robot, reducing the heterogeneous multi-agent system to a homogeneous network of double integrators on which standard distributed control theory could be applied.

Extensive MATLAB/Simulink simulations over nine scenarios demonstrated the effectiveness and robustness of the proposed architecture. In the nominal case, the three robots successfully converged to the desired equilateral triangular formation while tracking a circular reference trajectory, with formation errors decaying to zero under both chain and star communication topologies. The two topologies produced comparable performance, consistent with the theoretical requirement that a directed spanning tree rooted at the leader is sufficient for formation convergence.

The analysis of additional scenarios revealed several important properties of the closed-loop system. First, the controller retained its effectiveness when applied directly to the nonlinear plant without feedback linearization, owing to the near-identity structure of the decoupling matrices B_i and the dissipative nature of the drift terms ϕ_i . Second, actuator saturation on the physical inputs introduced a persistent steady-state error and a degraded transient, but did not prevent formation convergence. Third, the controller proved robust to imperfect model knowledge in the feedback-linearizing law, with negligible performance degradation compared to the nominal case. Fourth, the dynamic link failure scenario highlighted the fundamental limitation of the proposed architecture: loss of topological connectivity in the communication graph inevitably leads to loss of formation for the disconnected agents, as the controller has no built-in mechanism for topology recovery. Finally, the analysis of the asymmetric leader gain $c_\ell = 1.5c$ demonstrated that the closed-loop performance is largely insensitive to moderate variations of the leader gain around the coupling gain.

Further analysis can be conducted using different trajectories, graph topologies or by designing the controller selecting different parameters for matrices Q , R and the coupling gain c .